# Lab Report: Training dynamics of deep models

## 1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. Each experiment was repeated 5 times with different random seeds.

## 2 Results

Figure 1 shows the test accuracy of CNN by dataset.

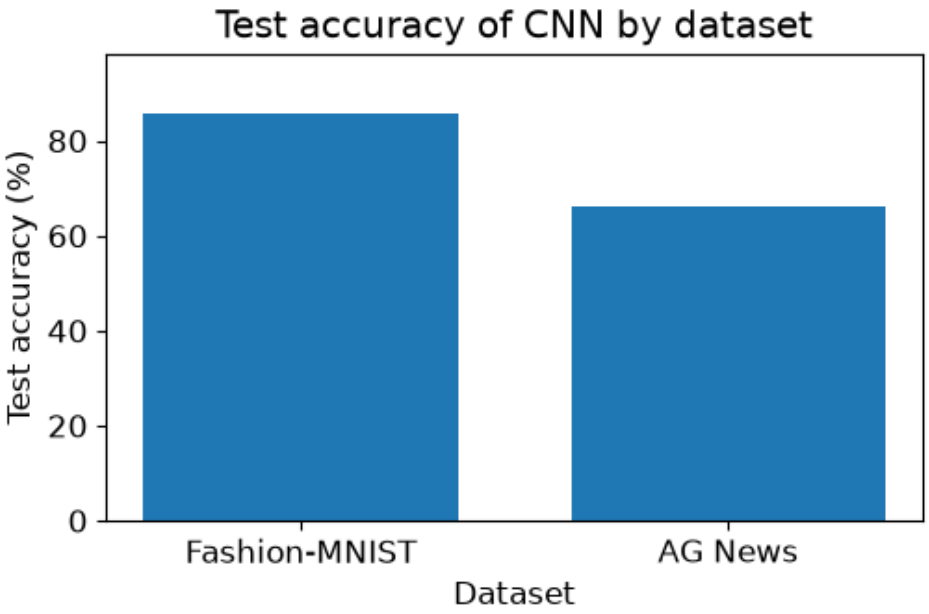



Figure 1: Test accuracy of CNN by dataset for the models in this experiment.

CNN scores 29.4 points higher on Fashion-MNIST than on AG News. As shown in Figure 1, CNN reaches over 80% test accuracy on Fashion-MNIST. CNN scores 39.3 points higher on Fashion-MNIST than on AG News (Figure 1).

Figure 2 shows the accuracy of DistilBERT by dataset.

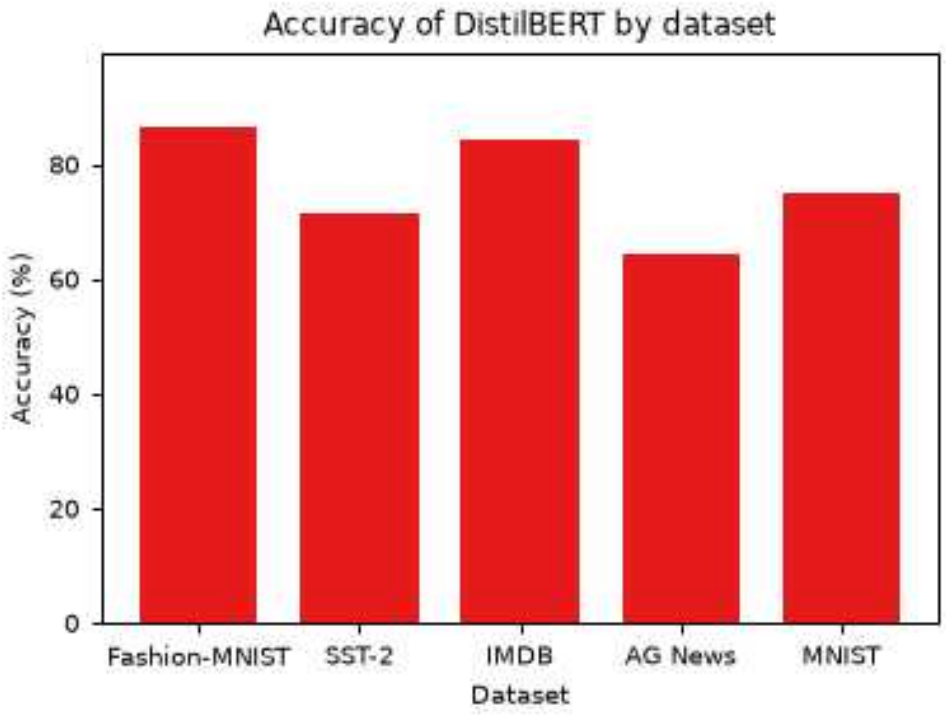


Figure 2: Accuracy of DistilBERT by dataset for the models in this experiment.

DistilBERT achieves 86% accuracy on Fashion-MNIST (Figure 2). DistilBERT reaches 84.4% accuracy on IMDB. DistilBERT scores 29.3 points higher on IMDB than on MNIST. As shown in Figure 2, DistilBERT reaches 86% accuracy on Fashion-MNIST.

Figure 3 shows the training loss.

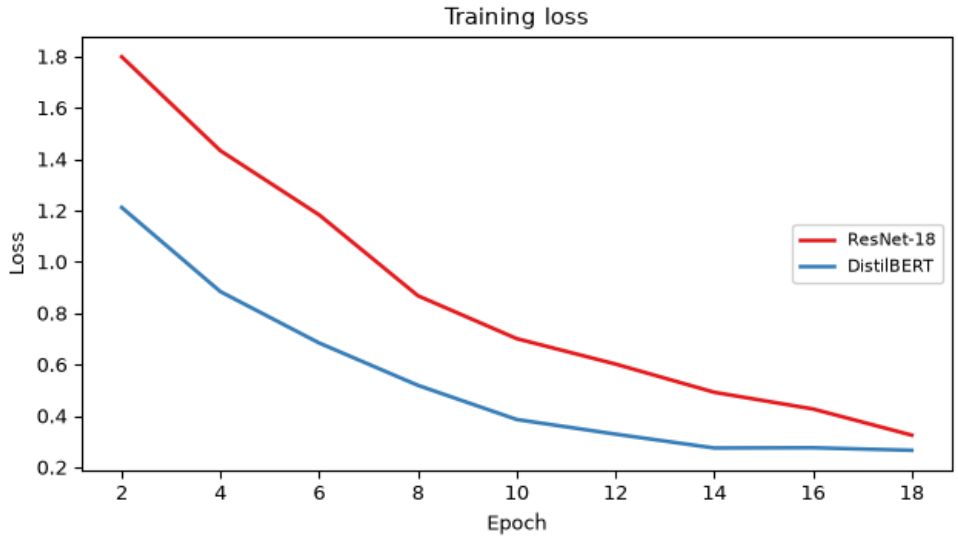



Figure 3: Training loss for the models in this experiment.

DistilBERT has a loss of 0.192 at epoch 16 (Figure 3). ResNet-18 has a loss of 0.266 at epoch 18 (Figure 3). As shown in Figure 3, ResNet-18 has a loss of 0.291 at epoch 18.

Figure 4 shows the accuracy by dataset.

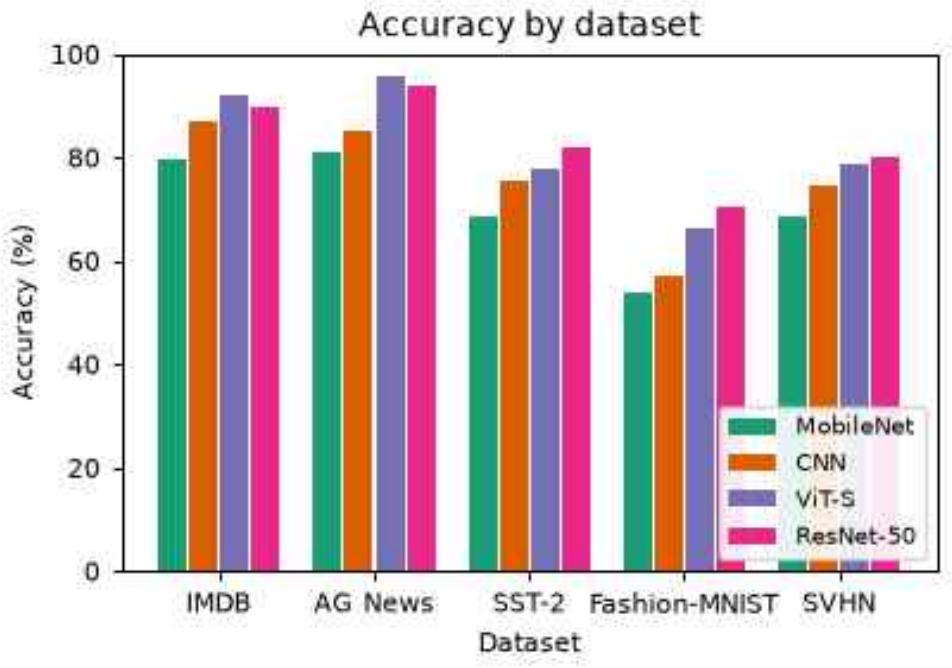



Figure 4: Accuracy by dataset for the models in this experiment.

On AG News, ResNet-50 is 28.4 points better than CNN. CNN reaches 65.7% accuracy on SST-2. On IMDB, CNN is 7.2 points better than MobileNet. ViT-S outperforms ResNet-50 by about 2 percentage points on IMDB.

## 3 Discussion

On MNIST, SVM obtains an accuracy of 57.1%. The differences between the models are discussed in the next section.